

Tutorial 1: PSET1

Tuesday, September 24, 2024

07:24 PM

Q1. (a)

① $\underline{100}$ ② $\underline{\underline{11000}}$ ③ $\underline{101}$
 Σ $\Sigma\Sigma \cdot \Sigma\Sigma$ Σ

③ $\underline{000101}$
 $\Sigma\Sigma\Sigma$

④ $\underline{10}$
 Σ

① no sigma, 10, sigma

② sigma, 10

③ sigma, sigma, 10, Σ , Σ

④ 3 sigmas

① $\underline{\quad} 10, \Sigma \underline{\quad}$
 $\downarrow \quad \downarrow$
 $\div 3 \quad \div 3$

② $\underline{\quad} \underline{\quad} 10 \underline{\quad}$
 $\div 3 \quad \Sigma \quad \div 3$

③ $\underline{\quad} \underline{\quad} \underline{\quad} 10 \underline{\quad} \underline{\quad} \underline{\quad}$

$\underbrace{\quad}_{\div 3} \underbrace{\quad}_{\Sigma} \underbrace{\quad}_{\Sigma} 10 \underbrace{\quad}_{\Sigma} \underbrace{\quad}_{\Sigma} \underbrace{\quad}_{\div 3}$

① (a). All strings of length odd such that 101 is a substring.

$\Sigma = \{0, 1\}$

$\underbrace{\quad}_{\downarrow} 101 \underbrace{\quad}_{\downarrow} \quad \underbrace{0}_{\downarrow} \underbrace{1}_{\downarrow} \underbrace{0}_{\downarrow} \underbrace{0}_{\downarrow}$
 $\quad \quad \quad 1, 3, \quad 1$

$\underline{00} 101 \underline{00}$

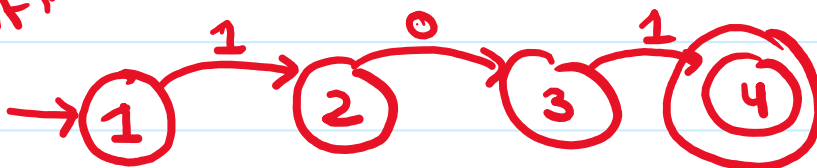
$(\Sigma\Sigma)^* 101 (\Sigma\Sigma)^*$ ✓

What is not correct =

$\Sigma^* 101 \Sigma^*$

$\downarrow$
 epsilon $\rightarrow 1011$ X

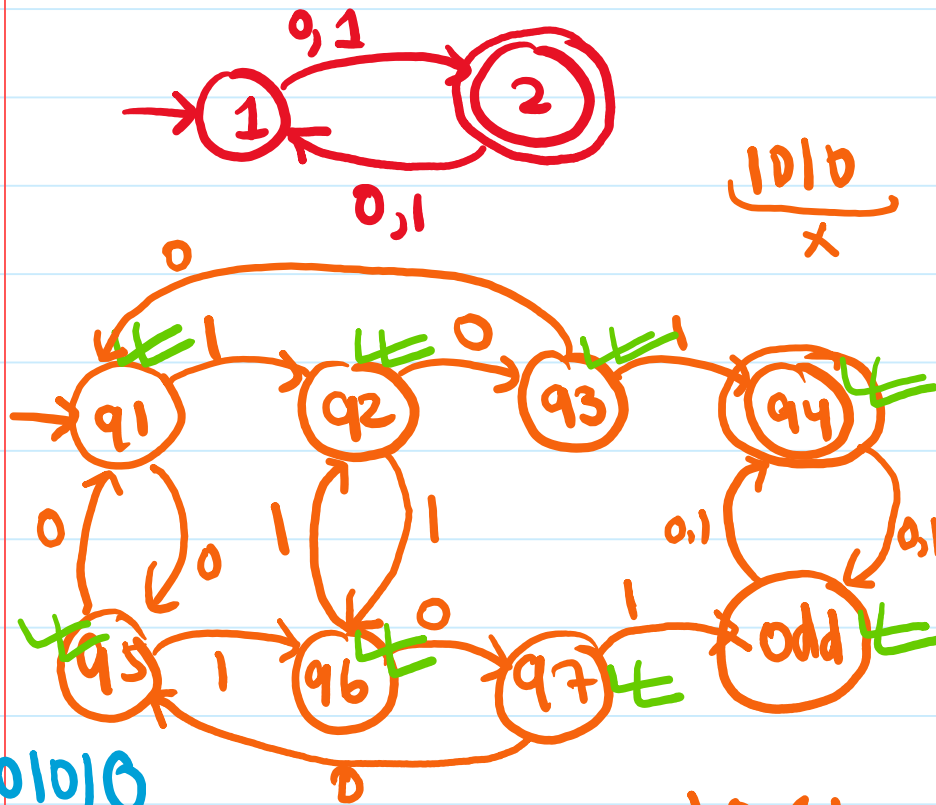
NFA



for substrings of 101

for substrings of 101

NFA/DFA



0001010

11101 → 5 ✓
→ 101 ✓

100101 1000101 100
10 10001010

RegEX : 11011

$[(\Sigma\Sigma)^* 101 (\Sigma\Sigma)^*] \cup$

$[\Sigma(\Sigma\Sigma)^* 101 \Sigma(\Sigma\Sigma)^*]$

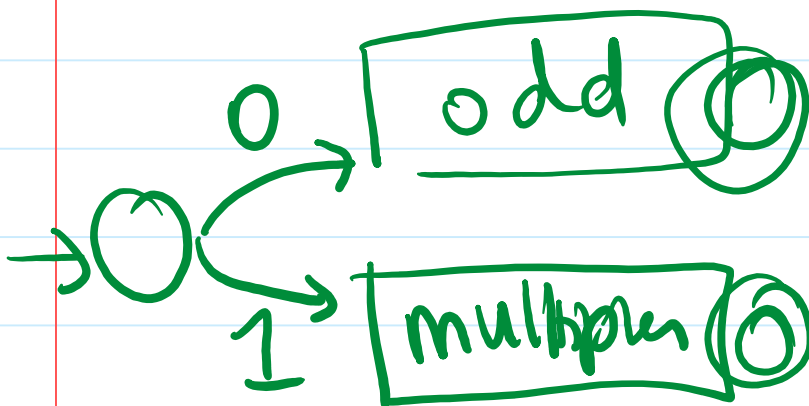
$\underbrace{(\Sigma(\Sigma\Sigma)^*)}_{\text{odd}} 101 \underbrace{\Sigma(\Sigma\Sigma)^*}_{\text{odd}}$

Q1(b)

$\underbrace{1011}_{\text{even}}$

$\Sigma \underbrace{101011}_{\text{odd}} XX$

$\underbrace{001}_{(\Sigma\Sigma)^4} \underbrace{101}_{\text{odd}} \underbrace{1}_{\Sigma} \underbrace{\Sigma\Sigma^*}_{\Sigma\Sigma^*}$



Q2(b)

$1^* 0^* 1^+ \rightarrow 1(1)^+$

Bonus
 (1) → pattern

strings

① → pattern

② → RE

abc ϵ

abc, bc, c, ϵ